PyData

Data Design Practices for EHR Data

Building trusted, standardized, and scalable healthcare data platforms

Preet Mehta — PyData Boston, November 4, 2026



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- Manager, Healthcare Analytics, EXL Service
- I love building and solving healthcare problems using Spark
- Open-source contributor — PyHealth, OpenTelemetry / CNCF (Inspektor Gadget, opentelemetry-python-contrib), and my own HIPAA pipeline framework
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The EHR Data Challenge: A Day in Mary's Care Journey

Multiple people, multiple departments, and continuously changing information — all for one patient.



Meet Mary

👤 85 years old

📍 Arrives at ABC General Hospital, Boston

📋 For a clinic visit

🕐 9:00 AM

👩‍💼

Administrative Staff

- Register Mary
- Verify insurance
- Capture demographics

🕐 9:30 AM

👩‍⚕️

Nurse

- Record vital signs
- – Blood pressure
- – Temperature
- – Pulse
- – Other observations

🕐 10:00 AM

👨‍⚕️

Doctor

- Examine Mary
- Document symptoms
- Write diagnosis
- Create treatment plan
- Order blood & urine tests

🕐 10:30 AM

🔬

Laboratory Technician

- Collect blood & urine samples
- Run laboratory tests
- Enter test results in the system

🕐 11:30 AM

👨‍⚕️

Doctor (Review)

- Review test results
- Update diagnosis and treatment plan
- Send prescription to pharmacy

🕐 12:00 PM

💊

Pharmacist

- Receive medication order
- Verify allergies & interactions
- Prescribe dosage
- Dispense medication to Mary

🕐 12:30 PM

👩‍💼

Administrative Staff

- Finalize visit and claims
- Update insurance details
- Schedule next appointment
- Complete discharge

🔄 Key Takeaways from Mary's Visit

⌛ **Information is continuously changing.**
A patient's record evolves throughout the visit.

⚠️ **It has real-world consequences.**
Missing allergy, wrong lab result, or incorrect dosage can impact clinical decisions and patient safety.

👥 **Different departments need to work together.**
Doctors need lab results, pharmacy needs medication orders, and nurses need the treatment plan.

🎯 What Can Go Wrong Without a Connected System?

💊 **Allergy not visible to the doctor**
→ Risk of prescribing a harmful medication.

🧪 **Lab result linked to the wrong patient**
→ Leads to incorrect diagnosis or treatment.

📋 **Incorrect dosage recorded by the nurse**
→ Can cause an underdose or overdose.

🖥️ That's Where an EHR Helps

✔ Brings all patient information together across departments.

✔ Keeps records up to date in real time.

✔ Ensures the right information is available to the right care team, at the right time.

✔ Supports safer, more coordinated, and higher-quality care for patients like Mary.

🖥️ Bottom Line

Healthcare data is not just a set of transactions — it's a connected, evolving record about a person's health, which requires an Electronic Health Record (EHR) to manage it safely and effectively.

Care Workflow: Before EHR vs. With EHR

From paper, faxes, and siloed records to a connected, real-time patient record

Before EHR: A Paper-Based, Siloed Process



Doctors

- Write clinical notes by hand
- Record diagnosis and plan
- Update paper chart



Nurses

- Manually record vital signs
- Chart medications given
- Use paper flow sheets





Laboratory

- Run lab tests
- Print results on paper
- Send results via fax or courier



Pharmacy

- Receive handwritten prescriptions
- Dispense medication
- Manually update records

Challenges without an EHR

- Fragmented information
- Delays in results
- Higher risk of errors
- No single, complete view of the patient

With EHR: A Connected, Real-Time Workflow



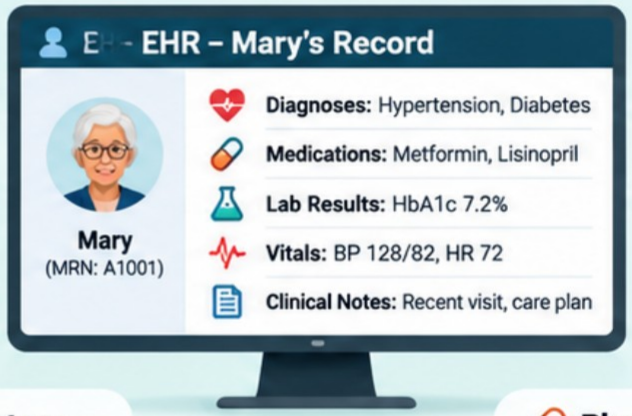
Doctors

- Enter diagnoses and care plan in the EHR
- View labs, medications, and past history



Nurses

- Record vital signs directly in the EHR
- Document assessments and medication administration





Laboratory

- Upload lab results directly to the EHR
- Results available instantly to the care team



Pharmacy

- Receive electronic prescriptions (eRx)
- Dispense and update medication records in the EHR

Benefits with an EHR

- One complete patient record
- Real-time information
- Fewer errors
- Safer, more coordinated care

Understanding EHR Architecture: One Patient, Many Connected Parts

An EHR organizes a patient's health information across time, visits, and care settings.

Where Does the Information Come From?

- **Hospitals**
  Admissions, diagnoses, treatments
- **Clinics**
  Consultations, follow-ups, care plans
- **Laboratories**
  Blood tests, urine tests, test results
- **Pharmacies**
  Medication orders, dispensing details
- **Radiology Centers**
  X-rays, CT scans, imaging reports
- **Telemedicine**
  Virtual consultations, clinical notes
- **Patient Mobile Apps**
  Health tracking, patient-reported data



Meet Mary

- 85 years old
- Multiple care visits over time
- Hypertension, Type 2 Diabetes

Mary's Care Journey (Longitudinal View)

2022 — **Hospital A**
Treated for hypertension

2023 — **Hospital B**
Managed for diabetes

2024 — **Laboratory**
Routine blood tests (e.g., HbA1c, cholesterol)

2025 — **Specialist Visit**
Medication review and follow-up

Today — **ABC General Hospital**
Current visit for ongoing care

How an EHR Organizes the Information?

Patient – Mary

The person whose health information is maintained (e.g., demographics, medical history, allergies)

Encounter – One visit or stay

A specific interaction with a healthcare provider (e.g., today's clinic visit, past admission, follow-up)

Clinical Records Within an Encounter

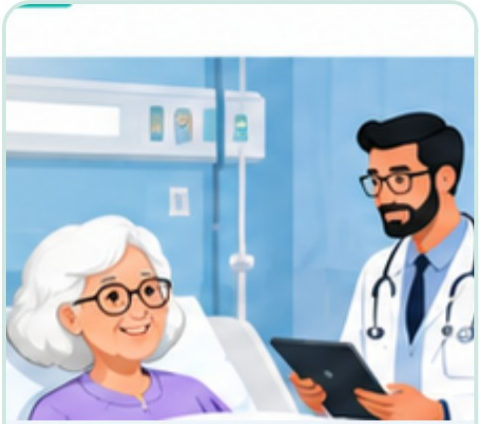
- **Diagnosis**
  Medical condition (e.g., Hypertension)
- **Medication**
  Prescribed medicine (e.g., Metformin)
- **Observation**
  Measurements & results (e.g., BP, HbA1c)
- **Procedure**
  Clinical intervention (e.g., Biopsy, Surgery)
- **Allergies**
  Known reactions (e.g., Penicillin – Rash)
- **Clinical Notes**
  Provider's assessment and care plan

Key Takeaway

One patient → Multiple encounters → Multiple clinical records, all connected over time to provide a complete view of the patient's health.

Database Architecture: Care First, Analysis Second

Designing healthcare data systems to support real-time patient care and large-scale analysis.



Mary's Visit
Care in Action

🖥️ OLTP – Operational System

Supports real-time patient care

- Record and update patient information
- Examples: vital signs, diagnosis, lab results, medication orders
- Built for speed, accuracy, and reliability

👥 Powers Day-to-Day Care

📊 OLAP – Analytical System

Used for reporting and analysis

- Analyze large volumes of historical data
- Examples: disease trends, readmissions, medication use, quality measures
- Built for insights and decision-making

📊 Enables Reporting, Analytics & AI

🛢️ How Healthcare Information Is Stored

🗂️ Relational Database (Structured Data)

Organizes clinical data in connected tables.

👤 Patient (Mary)

🗓️ Encounter (Visits)	🩺 Diagnosis (Conditions)	💊 Medication (Prescriptions)	📈 Observation (Vitals / Labs)

📄 Document Storage (Unstructured Data)

Manages flexible, text-heavy information.

- Clinical notes (doctor's notes)
- Discharge summaries
- Radiology reports (X-ray, CT, MRI)
- Referral and other scanned documents

🛡️ ACID: Ensuring Reliable & Safe Clinical Data

⚙️ Atomicity
All changes happen completely, or not at all.

🛡️ Consistency
Data follows defined healthcare rules.

🗂️ Isolation
Multiple updates don't interfere with each other.

☁️ Durability
Once saved, information remains secure and available.

🎯 Key Takeaway

Use the right system for the right purpose — keep patient care fast and reliable, while enabling large-scale analysis for insights and better healthcare decisions.

Healthcare Interoperability: How Hospitals Share Patient Information

Enabling different healthcare systems to exchange and use patient information in a secure, standardized way.

📍 Mary's Journey Across Multiple Hospitals



Mary
85 years old

🏥 Hospital A

Treated for Hypertension

🩺 **Diagnosis:** Hypertension

💊 **Medication:** Lisinopril

🧪 **Key Result:** BP 140/90

🏥 Hospital B

Treated for Diabetes

🩺 **Diagnosis:** Type 2 Diabetes

💊 **Medication:** Metformin

🧪 **Key Result:** HbA1c 7.2%

🏥 ABC General Hospital (Boston)

Current visit

✔ Doctor reviews Mary's history from Hospital A & B to make an informed treatment plan.

☁️ Health Information Exchange (HIE)

Securely connects hospitals to share authorized patient information.

🪪 Patient Information	📋 Diagnoses (Conditions)	💊 Medications (Prescriptions)	🧪 Lab Results (e.g., HbA1c, Cholesterol)	📄 Clinical Notes (Visit Summary)

⚙ Key Interoperability Building Blocks

🔗 Health Information Exchange (HIE)

A secure network that connects multiple healthcare organizations to share authorized patient information.

💬 HL7 Messaging

A widely used standard to exchange clinical events, such as admissions, discharges, transfers, orders, and test results.

🖥 FHIR (Fast Healthcare Interoperability Resources)

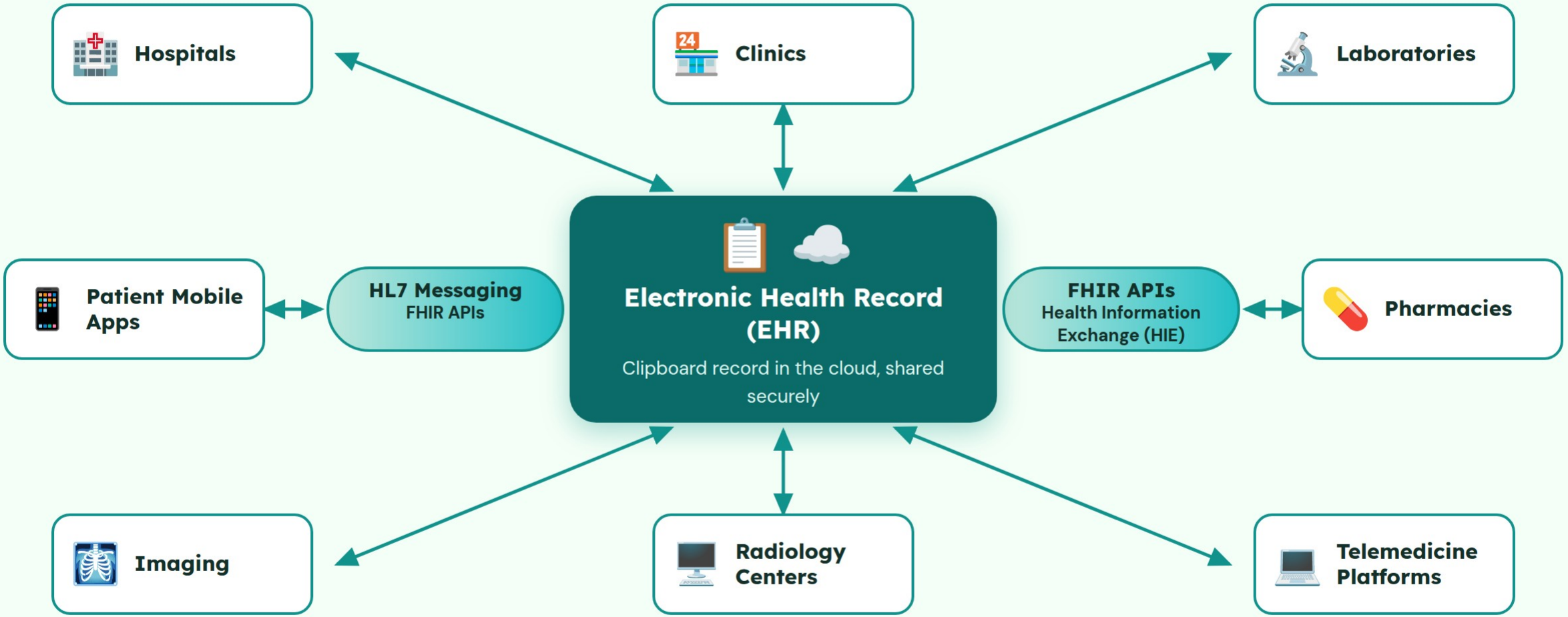
A modern standard that uses APIs to exchange structured clinical information (e.g., patient, medications, lab results).

🎯 Key Takeaway

An EHR helps manage patient information within a healthcare organization, while interoperability enables that information to be exchanged across different organizations.

The EHR ecosystem: connected by HL7, FHIR, and HIE

One central record, many connected systems



Standardized Data Modeling: Making Healthcare Data Speak the Same Language

A common structure and meaning for healthcare data, so information from different hospitals can be used reliably.

⚠ The Challenge: Same Information, Different Formats

Different hospitals may capture the same patient information in different ways.

Hospital A
DOB: 1944-05-12
Diagnosis: Hypertension
Code: I10 (ICD-10)

Hospital B
Birth_Date: 05/12/1944
Diagnosis: High Blood Pressure
Code: 38341003 (SNOMED CT)

ABC General Hospital
DateOfBirth: 12-May-1944
Diagnosis: Hypertension
Code: 59621000 (SNOMED CT)



Same patient (Mary), same condition, but captured differently across systems.

⚙ Key Building Blocks for Standardized Healthcare Data

1 FHIR Resources

A standard set of building blocks to represent healthcare information.

- **Patient** — Demographics (e.g., Mary, 85)
- **Encounter** — A visit or stay (e.g., Hospital A)
- **Condition** — A diagnosis (e.g., Hypertension)
- **Observation** — A result or measure (e.g., BP, HbA1c)

2 Clinical Terminologies

Standard vocabularies that give clinical concepts a consistent meaning.

- **ICD-10** — Diagnoses & disease classification
- **SNOMED CT** — Detailed clinical concepts
- **LOINC** — Laboratory tests & observations
- **RxNorm** — Medications & drug information

3 Canonical Data Model

A common, standardized structure to integrate data from multiple hospitals.

Data Element	Hospital A	Hospital B	Hospital C	Standard Model
Patient ID	PATIENT_ID	Patient_Number	PAT_ID	**patient_id**
Patient Name	FIRST_NAME + LAST_NAME	Full_Name	PATIENT_NAME	**patient_name**
Date of Birth	DOB	Birth_Date	DATE_OF_BIRTH	**date_of_birth**

4 Data Validation

Ensures the data is accurate, complete, and clinically meaningful before it is used.

- ✓ Valid and consistent formats (e.g., dates, units)
- ✓ Correct and mappable clinical codes (e.g., ICD-10, LOINC)
- ✓ Required information is not missing
- ✓ Logical relationships are maintained (e.g., lab result linked to the right patient and encounter)

◎ Key Takeaway

Standardized data modeling brings consistency in structure and meaning, so healthcare information from different hospitals can be integrated, understood, and used reliably.

Multi-EHR Data Integration: Three Hospitals, Three Different Schemas

Bringing together patient information from different EHR systems into one unified, consistent view.



Mary
85 years old

🗓️ Care Across Three Hospitals

- ✔ **Hypertension** (Hospital A)
- ✔ **Diabetes** (Hospital B)
- ✔ **Current visit** (ABC General)

🛢️ Same Patient, Different EHR Structures

🏥 Hospital A
(EHR System A)

🪪 **first_name**	Mary
🪪 **last_name**	Sharma
🗓️ **dob**	1944-05-12
👤 **patient_id**	A100

🏥 Hospital B
(EHR System B)

🪪 **full_name**	Mary Sharma
🗓️ **birth_date**	05/12/1944
👤 **patient_number**	B250
💊 **diagnosis**	E11 (Diabetes)

🏥 ABC General Hospital
(EHR System C)

🪪 **patient_name**	M. Sharma
🗓️ **date_of_birth**	12-May-1944
👤 **pat_id**	C789
💊 **diagnosis**	I10 (Hypertension)

➜

🗂️ Unified (Canonical) Patient Model

Standard Field	Unified Record (Mary)
👤 **Enterprise Patient ID**	P10001
👥 **Patient Name**	Mary Sharma
🗓️ **Date of Birth**	1944-05-12
🔗 **Source Patient IDs**	A100, B250, C789
🩺 **Conditions (Mapped Codes)**	Hypertension (I10), Diabetes (E11)

⚙ Key Integration Challenges & How We Address Them

🗂️ 1. Column Mappings

Different hospitals use different field names, formats, and structures.

first_name (Hospital A) → **full_name** (Hospital B) → **patient_name** (Standard)

🔑 2. Primary Keys Do Not Travel

Patient IDs are unique within each hospital and cannot be directly used across hospitals.

👤 **A100** (Hospital A) ≠ 👤 **B250** (Hospital B)

ℹ️ Different IDs don't mean different people.

👥 3. Patient Identity Resolution

Determine records from different hospitals that belong to the same individual.

- ✔ Match using multiple attributes (e.g., name, DOB, etc.)
- ✔ Create a Master Patient Index (MPI) / linked patient record
- ✔ Preserve source IDs and match confidence

🎯 Key Takeaway

Integrate data from multiple EHRs by mapping to a common model, preserving source identifiers, and correctly identifying the same patient across systems.

Data Quality & Governance: Making Healthcare Data Trustworthy

Ensure patient data is accurate, complete, consistent, timely, and responsibly managed across its lifecycle.



Mary's data comes from multiple hospitals. We need to ensure it is complete, correct, current, and trusted.

Data Quality: Five Key Dimensions for Mary's Patient Record

1. Accuracy

Are the values and patient details correct?

Example: Correct blood pressure (128/82 mmHg)

2. Completeness

Are important clinical details missing?

Example: Missing allergy information

3. Consistency

Do different systems represent data in a consistent way?

Example: Same date of birth across hospitals



Mary's Unified Patient Record
(From Hospitals A, B and C)

4. Timeliness

Are updates arriving on time?

Example: Latest medication list after a recent visit

5. Uniqueness

Are duplicate patient, encounter, or result records being created?

Example: Preventing the same encounter from being loaded twice

Data Governance: The Rules & Oversight

Data Ownership
Who is responsible for each type of data (e.g., diagnoses, medications)?

Access Control
Who can access the data and for what purpose?

Data Definitions
A common understanding of key terms (e.g., what counts as a readmission).

Data Lineage
Trace the data back to its source (e.g., Hospital B, original record).

Data Retention
How long the data should be kept based on policies and regulations.

Key Takeaway **In healthcare, data integration tells us that information has arrived. Data quality and governance tell us whether we can trust it.**

Security and Privacy: Protecting Patient Information

Ensure the right people can access the right information, while keeping patient data private, secure, and compliant.

👥 The Scenario: Many People, Different Needs

Multiple hospital teams access Mary's information to provide care, but each needs only the information relevant to their role.



Mary
85 years old

Doctor
Review medical history, diagnosis, and treatment plan

Nurse
Record vital signs (e.g., blood pressure, temperature)

Lab Technician
Access test orders and record results

Pharmacist
Review medications and allergies

Key Security Practices in Healthcare

1. Role-Based Access Control (RBAC)
Grant access based on job responsibilities.

Example: Doctors see clinical records, pharmacists see medication details, and admins see registration and insurance info.

2. Data Encryption
Protect patient data when it is stored and when it moves between systems.

Example: Encrypted storage and secure data exchange between hospitals.

3. Audit Trails
Track who accessed a patient's record, when, and what actions they performed.

Example: Log when a user views Mary's medical record.

4. Patient Privacy & Data Protection
Use and share patient information only for permitted purposes and in line with applicable regulations.

Example: Share Mary's records with another hospital only with proper authorization and safeguards.

🎯 Key Takeaway **The right information, to the right person, at the right time — without compromising patient privacy or security.**

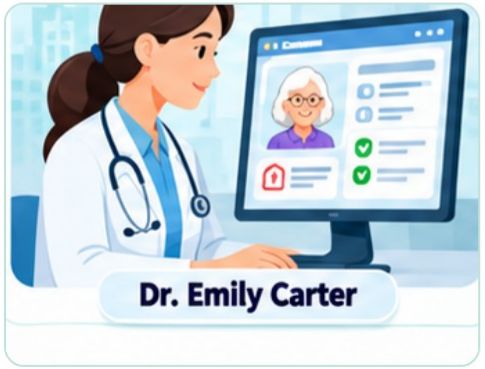
Clinical Usability: Designing EHRs for the People Who Use Them

Turn healthcare data into clear, relevant, and actionable information for better and faster patient care.

👥 The Clinical Scenario: Mary's Visit

Dr. Emily Carter is reviewing Mary's record before her visit. She needs to quickly understand:

- ✓ Medical conditions (e.g., hypertension, diabetes)
- ✓ Current medications
- ✓ Allergies
- ✓ Latest lab results (e.g., HbA1c)
- ✓ Recent hospital visits
- ✓ Previous treatment plan



🖥️ Key Elements of Clinical Usability

1. Clear Presentation of Information

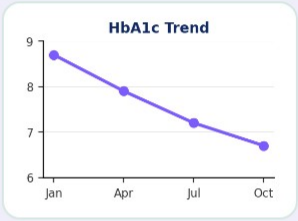
Organize key patient details in a simple, easy-to-find way.

Example: Quick view of allergies, medications, diagnoses, and latest lab results.

2. Meaningful Visualizations

Show trends and patterns to support better clinical decisions.

Example: HbA1c trend over the last 12 months to see if diabetes is improving.

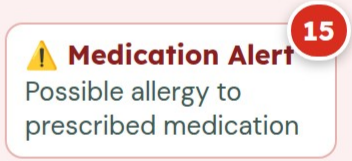


3. Relevant Alerts and Notifications

Alert Fatigue — When healthcare professionals get too many unnecessary alerts, they may start ignoring them — that is called alert fatigue.

Notify clinicians about important clinical risks, while avoiding unnecessary alerts.

Example: Warning for a potential medication–allergy conflict.



4. Structured and Unstructured Information

Support both standardized clinical data and detailed clinical narratives for a complete patient picture.

Example: Vitals and lab values (structured) + physician notes (unstructured).

✓ Structured

✓ Blood pressure 128/82 ✓ Heart rate 72 bpm ✓ Temperature 98.4°F ✓ Diagnosis code I10

📄 Unstructured

"Mary has been feeling unusually tired for the last two weeks, and sometimes feels dizzy."

🎯 Key Takeaway

An EHR is not successful just because it stores information — it is successful when healthcare professionals can find, understand, and use that information effectively to support care.

Building an Analytics-Ready Healthcare Data Platform

Turn clinical data from multiple systems into reliable, consistent, and meaningful insights for care, operations, and research.

👥 The Scenario: Mary's Complete View



Mary, 85 years old
has received care at multiple hospitals.

Hospital A (EHR A)
Treated for hypertension

Hospital B (EHR B)
Treated for diabetes

ABC General Hospital (EHR C)
Now treating her for another health concern

🗂️ The Journey: From Raw Data to Meaningful Insights

Bronze

Preserve What We Receive

- ✓ Collect data from multiple clinical systems (EHRs, labs, pharmacy, etc.)
- ✓ Keep the original information as received
- ✓ Track the source and when it arrived

→

Silver

Clean, Connect, and Standardize

- ✓ Link records for the same patient (e.g., Mary across hospitals)
- ✓ Standardize formats and clinical terms
- ✓ Remove duplicate records
- ✓ Validate and improve data consistency

→

Gold

Prepare for Specific Questions

- ✓ Organize data for reporting, analytics, and research
- ✓ Create trusted, business-ready datasets
- ✓ Answer key healthcare questions

🎯 What Can We Answer?

- 👥 How many patients have diabetes?
- How many patients were readmitted within 30 days?
- How have patients' HbA1c levels changed over time?
- 📋 Which patients may need additional follow-up?

🎯 Key Takeaway — **We preserve the data, make it consistent, and prepare it to answer meaningful healthcare questions.**

Practical Implementation: From Multiple EHRs to One Trusted Healthcare View

A step-by-step approach to bring clinical information from different hospitals together, making it consistent, accurate, and useful.

The Scenario: Mary's Journey



Mary, 85 years old has received care at multiple hospitals for different health conditions.

Hospital A

Treated for hypertension (high blood pressure)

Hospital B

Treated for diabetes (including HbA1c results and medications)

ABC General Hospital

Now treating her for another health concern and ongoing care

The Implementation Approach: 5 Key Steps

1 Identify Data Sources

Example: Hospital A, Hospital B, and ABC General Hospital

2 Understand and Organize

Example: Patient, encounter, diagnosis, medication, and lab results

3 Standardize and Connect

Example: Resolve different patient IDs, align date formats, and standardize clinical terms



4 Validate and Protect

Example: Check for missing or duplicate records, fix inconsistencies, and apply access controls

5 One Trusted Healthcare View

Example: A unified record that supports reporting, analytics, research, and better care decisions

The Outcome

- A more complete patient history
- Better and more informed care decisions
- Meaningful reporting (e.g., readmissions, chronic condition trends)
- Support for research and population health
- Secure and compliant use of patient information

Key Takeaway

Integration is not just moving data. It's about understanding, standardizing, validating, and connecting information to create one trusted view of the patient.

Key Takeaways: Designing Reliable and Useful Healthcare Data

The core principles to turn healthcare information into better care, better insights, and better outcomes.



Mary's Journey — The Big Picture

A connected, accurate, secure, and meaningful view of her health across multiple hospitals.



EHR Architecture

Brings together information from doctors, nurses, labs, and pharmacies.

Database Architecture

Provides reliable, fast, and consistent storage for clinical information.

Interoperability

Helps hospitals securely share patient information (e.g., Mary's records).

Standardized Data Modeling

Uses common definitions (e.g., patient, diagnosis, medication) across hospitals.

Multi-EHR Integration

Brings information from different systems together into a unified view.

Data Quality

Ensures the information is complete, accurate, consistent, and free of duplicates.

Governance & Privacy

Controls who can access patient information and keeps it secure and compliant (e.g., HIPAA).

Analytics-Ready Platform

Transforms raw clinical data into trusted information for reporting, research, and better care decisions.

🎯 The Bottom Line | **Good healthcare data design is not just about storing information — it's about making it connected, trusted, secure, and useful to improve patient care.**

Thank You Q&A

Preet Mehta

LinkedIn

Slides Explained

Slides Explained — Scan for a detailed, slide-by-slide explanation of this talk, the slides download, my newsletter, and ways to connect, book a call, or follow me on social media.